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Session Date: April 28, 2026 (1 hour)

Objective

Get the graph loading working using CSR format.

Attempts made

I wasn't sure how to build CSR without knowing all the out-degrees first. My first attempt just used a map to collect adjacency lists on the fly and then tried to convert it after:

```
cppmap<int, vector<int>> adj;

while (getline(fin, line)) {

    int u, v;

    iss >> u >> v;

    adj[u].push_back(v);

}
```

It worked on tiny inputs but felt wrong and was going to be slow on a large graph. Also it kept segfaulting on the comment lines in the dataset because I wasn't skipping them and iss >> u was reading garbage.

Solutions/Workarounds

Figured out I needed two passes. First pass just reads all the edges into a vector and finds the max node ID. Second pass counts out-degrees, does a prefix sum to get the offsets, then fills the edges array using a position cursor:

```
cppfor (auto& [u, v] : edge_list) offsets[u + 1]++;

for (int i = 1; i <= num_vertices; i++) offsets[i] += offsets[i-1];

for (auto& [u, v] : edge_list) edges[pos[u]++] = v;
```

Fixed the crash by just skipping comment lines at the top of the loop:

```
cppif (line.empty() || line[0] == '#') continue;
```

Tested on the small example from the spec and checked offsets and edges by hand. Then ran on the full dataset and got:

Graph loaded: 82168 vertices, 948464 edges

Commits: Add CSR graph loader from edge list file

Experiments/Analysis

None this session.

Learning outcome

The prefix sum is what makes CSR work, once you have it, finding any vertex's neighbors is just indexing into a flat array. I kept trying to avoid the two passes but there's no way around it since you need all the degrees before you can place anything.